Same parameters, Adam moments and frozen supervision within each column
64 complete trajectories per update; FP32, eta=0.3, Top-16 production gradient

Parent update 40

Parent update 120

Parent update 240

One-step full-BPO reduction (%)

LR / 1e-6

Full BPO / branch initial

Branch optimizer updates

LR
1.25e-07
2.5e-07
5e-07
1e-06
2e-06
4e-06
8e-06
1.6e-05

Pre-clip full-pool gradient norm